Madhan Senthilkumar

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PROFESSIONAL SUMMARY

Data Scientist with 2 internships and 4 projects applying statistical analysis, machine learning, deep learning, and LLM-based systems to real-world problems. Built FitCheck, a dual-mode Explainable RAG platform for resume-JD matching and candidate ranking. Built a churn risk-scoring model (81.5% accuracy) with SHAP explainability, and an end-to-end computer vision system (YOLOv8 + ByteTrack) for real-time vehicle detection and violation tracking. Proficient in Python, SQL, Pandas, Scikit-learn, LangChain, and OpenCV across the full pipeline: EDA, feature engineering, model building, evaluation, and interpretability. IBM-certified in Python for Data Science.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning — Parul University, Vadodara   CGPA: 8.30 | Aug 2023 – 2027

EXPERIENCE

Data Science Intern   *Future Interns | Dec 2025 – Jan 2026*

- Engineered a weighted KPI scoring model across 5-point Likert scale survey data to quantify student satisfaction, surfacing a top-performing metric at 4.21/5 and flagging 3 critical areas below 3.6/5 for institutional review.
- Translated raw survey responses into an interactive visual report using Pandas and Seaborn — bar charts and sentiment distribution charts — generating actionable recommendations for 2 key improvement areas.

Machine Learning Intern   *Cognifyz Technologies (Remote) | Nov 2025 – Dec 2025*

- Designed and executed end-to-end EDA and preprocessing pipelines using Python and Pandas on large-scale datasets, improving data reliability and quality for downstream modeling.
- Built, tuned, and evaluated multiple classification and regression models using Scikit-learn; selected the best-performing model based on accuracy, precision, and F1-score metrics.

PROJECTS

FitCheck – Dual-Mode Explainable RAG Platform   [GitHub]   [Live Demo]   *React, FastAPI, LangChain, ChromaDB, Groq (Llama 3.1) | 2026*

- Built a dual-mode RAG platform enabling candidates to detect cross-JD requirement conflicts and recruiters to auto-rank up to 10 resumes via a source-cited, confidence-scored matching engine.
- Engineered structured LLM-based candidate scoring (match %, why-select/why-not-select rationale) with a fairness-audit layer disclosing sensitive-signal presence without influencing rankings.

Customer Churn Prediction & Retention Platform   [Live Demo]   *Python, Streamlit, Scikit-learn, SHAP, Plotly | 2026*

- Applied SHAP feature attribution to explain per-customer churn drivers (contract type, tenure, payment method), turning a black-box classifier into an actionable retention tool.
- Built a real-time churn risk-scoring model (81.5% accuracy) segmenting 7,000+ Telco customers into Low/Moderate/High risk tiers with live confidence scoring.

AI-Powered Traffic Analytics System   [GitHub]   *Python, YOLOv8, ByteTrack, OpenCV, Streamlit, SQLite | 2026*

- Built an end-to-end CV pipeline with YOLOv8 detection and ByteTrack tracking to persistently identify vehicles/pedestrians across live video frames.
- Engineered speed estimation and traffic density classification (Low/Medium/Heavy), with rule-based violation detection logged to SQLite and visualized on a live Streamlit dashboard.

RetailAnalytics – Sales Intelligence Platform   [GitHub]   *Python, FastAPI, SQL, Scikit-learn, Prophet | 2026*

- Engineered RFM features via SQL and built a K-Means model to segment customers into 3 behavioral clusters for targeted retention strategy.
- Combined Isolation Forest anomaly detection with Prophet time-series forecasting to flag pricing outliers and predict daily sales (evaluated via MAE, RMSE, R^2).

TECHNICAL SKILLS

Generative AI & LLM: LangChain, RAG, Vector Databases (ChromaDB), Groq API, Prompt Engineering, Sentence-Transformers
Machine Learning: Scikit-learn, Classification, Regression, Clustering, Time-Series Forecasting, Anomaly Detection, SHAP
Deep Learning & Computer Vision: TensorFlow, Keras, YOLOv8, OpenCV, Object Detection, Multi-Object Tracking (ByteTrack)
Statistics: Descriptive Statistics, Hypothesis Testing, Correlation Analysis
Programming & Querying: Python, SQL, Java, Aggregations, Joins
Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering
Data Visualization: Matplotlib, Seaborn, Plotly, Power BI
Database: MySQL, MongoDB, SQLite
Tools & Platforms: FastAPI, React, Jupyter Notebook, Google Colab, Git, GitHub, VS Code, Docker, Streamlit

CERTIFICATIONS & ACHIEVEMENTS

- IBM – Python for Data Science (2025)
- Oracle – Agentic AI Certified Foundations Associate
- HackerRank – SQL (Advanced) Certificate
- Deloitte Australia – Data Analytics Job Simulation (2024)
- Analytics Vidhya – Analyzing and Visualizing Data with Microsoft Power BI